

Chapter 3

Hilbert Space Topology

In previous lessons, several essential notions were used intuitively without being formally defined: what is a convergent sequence? what do we mean by continuity of a linear map? how can a dense subspace be precisely characterized? These questions belong to *topology*, the branch of mathematics that formalizes the notions of proximity, limits, and continuity. This lesson provides an introduction.

3.1 General topological spaces

A topology on an arbitrary set E is defined by a collection of open sets $\mathcal{T}$:

1. $\emptyset \in \mathcal{T}$ and $E \in \mathcal{T}$,
2. any arbitrary union of open sets is an open set,
3. any finite intersection of open sets is an open set.

The following paragraphs aim to provide an intuitive explanation for the notion of an open set. We first note (see below) that on $\mathbb{R}$, the archetypal example is the open interval $I =]0, 1[$. One observes that within this interval, every point x always has a surrounding region entirely contained in I , for instance $]x - \varepsilon, x + \varepsilon[$ for ε small enough. This is a kind of "mobility zone" in which one can move while remaining "close" to x and without leaving the set I . The open set $]0, 1[$ is therefore a set *without boundary*: one can approach its boundary (0 or 1) arbitrarily closely without ever reaching it.

A topology thus defines a notion of *locality* around each point: the open sets containing x describe the "zones" in which x may be situated. The more open sets are available, the more points can be *separated*: if $x \neq y$ and there exists an open set containing x but not y , then the topology "sees" that these two points are distinct. Conversely, if every open set containing x also contains y , then x and y are topologically indistinguishable. It follows that the more open sets a topology $\mathcal{T}$ contains, the greater its separating power. Such a topology is said to be *finer*.

Note that this structure requires no notion of distance: at this stage one cannot say that one point is *closer* to x than another, only that it shares or does not share certain open sets with it. In this sense, a topology is a weaker structure than that of a metric space, where the distance provides a quantitative measure of proximity via open balls $B(x, \varepsilon)$ of arbitrarily small radius (see below).

With this in mind, the definition of a topology becomes intuitively clear. Stability under finite intersections is natural: it guarantees that the mobility zone around x retains a positive "thickness" even as it shrinks under intersections. Allowing infinite intersections would allow one to construct intersections that are no longer open—typically singletons $\{x\}$ —which would collapse the structure: one would lose the notion of continuous motion and proximity¹.

Stability under arbitrary unions of open sets reflects the extensible nature of locality: if a point belongs to each of these mobility zones, then their union is also a mobility zone for that point, forming a new open set containing it.

A topological space is by definition a set E equipped with such a topology $\mathcal{T}$.

3.2 Topology of normed spaces

In a normed vector space (NVS) E , the norm naturally generates a topology through the associated distance $d(x, y) = \|x - y\|$. The distance provides a concrete way to construct "mobility zones" in the form of open balls of radius $r > 0$ centered at a point x_0 , denoted $B(x_0, r)$:

$$B(x_0, r) = \{y \in E : \|y - x_0\| < r\}.$$

Definition 3.1 : Strong topology

On a NVS E , the **strong topology** (or *norm topology*) is the topology whose open sets are the subsets $U \subset E$ such that

$$\forall x \in U, \exists r > 0, B(x, r) \subset U.$$

In other words, U is open if every one of its points admits an open ball centered at it and entirely contained in U (see Fig. 3.1).

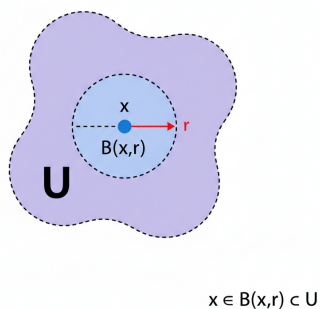


Figure 3.1: A set U is an open set of the strong topology if and only if each of its points x can be covered by an open ball contained in U .

¹This is not impossible in itself. Mathematically, the discrete topology on a set of points does contain all singletons in particular; but this is beside the point in our context, since we are dealing with "continuous" spaces and wish to exploit that structure.

It is straightforward to verify that this family of subsets satisfies the three axioms of a topological space. This topology admits the following equivalent, more constructive description:

Proposition 3.1 : Description by unions of balls

The open sets of the strong topology are exactly the arbitrary unions of open balls:

$$U \in \mathcal{T} \iff U = \bigcup_{i \in I} B(x_i, r_i)$$

for some family $(B(x_i, r_i))_{i \in I}$ of open balls, indexed by an arbitrary set I (finite, countable, or uncountable).

Remark: Throughout the rest of this course, we shall always work within a NVS or a Hilbert space. We will establish a number of *sequential characterizations* of topological properties (closure, continuity, density, etc.), meaning that purely topological properties will be expressed as statements about sequences. It is important to understand that these characterizations, which are often very practical, all arise from the same underlying mechanism in a NVS.

Specifically, to study the behavior of a subset or a map near a point x_0 , one exploits the availability of a countable family of nested balls $B(x_0, 1/n)$ whose radii tend to 0. Choosing a point $x_n \in B(x_0, 1/n)$ in each ball yields a sequence (x_n) converging to x_0 ; the topological property under study then translates into properties of these sequences.

This is also why the sequential characterizations stated in this lesson are specific to metric spaces—in particular to NVS and Hilbert spaces—and do not extend to general topological spaces.

3.3 Closed sets, closure, and dense subsets

We begin with two immediate applications of this sequential mechanism to the notions of *closed set* and *closure*. The counterpart of open sets is closed sets: a closed set is generally defined as a subset of E whose complement is open. In a normed space, this definition admits a very useful sequential characterization:

Theorem 3.1 : Sequential characterization of closed sets

Let E be a NVS and A a subset of E . Then A is a closed set if and only if for every sequence (x_n) of elements of A converging to a point $x \in E$, one has $x \in A$.

In other words, a closed set is stable under taking limits: limits of sequences of elements of the closed set remain in the closed set. Given any subset A of E , one can augment it with all limits of sequences of elements of A . This extension is called the *closure* of A , and² is denoted $\bar{A}$:

²In general topology, the closure of A is the smallest closed set containing A . The definition we give here is actually a proposition: it is the sequential characterization of the closure in a NVS, in accordance with the remark made above.

Definition 3.2 : Closure

A point $x \in E$ belongs to the closure $\bar{A}$ if and only if there exists a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of A converging to x .

It follows from these two facts that a set A is closed if and only if it equals its own closure $A = \bar{A}$: the set already contains all its limit points. We began this chapter with the open interval $I =]0, 1[$. One can easily construct sequences of elements of I that converge to 0 or to 1. Moreover, constant sequences $x_n = x \in I$ obviously converge to x . Consequently, the closure of this open set is the closed interval $[0, 1]$.

Remark: The sequential characterizations above involve the notion of convergence of a sequence, which will be formalized in the next section. For now, the reader may rely on the usual intuition: $x_n \rightarrow x$ means that the x_n approach x arbitrarily closely in the sense of the norm, $\|x_n - x\| \rightarrow 0$.

Dually to the closure, the *interior* of a subset A is defined as the largest open set contained in A , denoted $\overset{\circ}{A}$. Equivalently, it is the set of points $x \in A$ for which there exists an open ball centered at x entirely contained in A . An open set is its own interior.

Finally, the *boundary* of A is the set of points that belong to the closure but not to the interior:

$$\partial A = \bar{A} \setminus \overset{\circ}{A}.$$

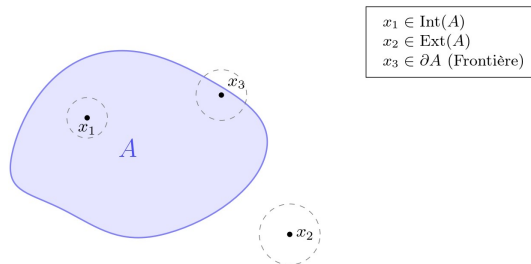


Figure 3.2: A subset A of E and three characteristic points: an interior point, a boundary point, and an exterior point. The dashed circles represent an open ball centered at each. The interior point admits a ball entirely contained in A ; the exterior point, a ball entirely disjoint from A ; the boundary point is such that every ball centered at it necessarily straddles both the interior and the exterior.

All these notions are particularly transparent in the case of the open ball of radius r centered at x_0 :

- The open ball is its own interior,
- The closure of the open ball is the closed ball, denoted $\overline{B(x_0, r)}$
- The boundary is the sphere of radius r centered at x_0 , denoted $S(x_0, r)$
- The exterior of the open ball, defined as the interior of its complement, equals $\{y \in E : \|y - x_0\| > r\}$.

The density of a subset A in E is a notion inherited from the preceding ones.

Definition 3.3 : Dense subset

A subset A of E is *dense* in E if its closure equals all of E : $\bar{A} = E$.

By Theorem 3.1, this means that A is dense in E if every point of E can be approximated arbitrarily closely by elements of A . For example, continuous functions are dense in $L^2([0, 1])$.

We note the following points:

- A dense subset cannot be closed, unless it equals the whole space. Indeed, if A is both dense and closed, then $E = \bar{A} = A$.
- A dense subset is not in general open: it must be able to approximate elements it does not contain. For example, $\mathbb{Q}$ is dense in $\mathbb{R}$ but is not open in $\mathbb{R}$: no open ball centered at a rational number is contained in $\mathbb{Q}$. On the other hand, $\mathbb{R} \setminus \{0\}$ is a dense open subset of $\mathbb{R}$: so it is possible, intuitively, by removing a "small" set of points from the whole space. More precisely, the theorem states that if F is a closed set with empty interior, then $E \setminus F$ is a dense open set.

3.4 Strong and weak convergence

The sequential characterizations we have just seen all rested on the intuitive notion of convergence $\|x_n - x\| \rightarrow 0$. It is time to formalize this. In a Hilbert space, we will moreover see that there exist *two* distinct notions of convergence: strong and weak.

3.4.1 Strong convergence of a sequence of vectors

In a NVS, a sequence (x_n) converges to x when its terms approach x arbitrarily closely.

Definition 3.4 : Strong convergence of a sequence of vectors

Let E be a NVS. A sequence $(x_n)_{n \in \mathbb{N}}$ of elements of E *converges strongly* to $x \in E$ if and only if

$$\|x_n - x\| \xrightarrow{n \rightarrow \infty} 0,$$

where this convergence in $\mathbb{R}$ is understood as

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \quad \|x_n - x\| < \varepsilon.$$

This is called *strong convergence* in reference to the strong topology, or equivalently *convergence in norm*. In Dirac notation, $|\psi_n\rangle$ converges strongly to $|\psi\rangle$ if and only if $\| |\psi_n\rangle - |\psi\rangle \| \rightarrow 0$.

Remark: Strong convergence implies convergence of norms. Indeed, by the reverse triangle inequality³:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| \xrightarrow{n \rightarrow \infty} 0$$

³The reverse triangle inequality on $\mathbb{R}$ states that for all real numbers a, b , $||a| - |b|| \leq |a - b|$. It follows from the usual triangle inequality applied to $a = (a - b) + b$, which gives $|a| \leq |a - b| + |b|$, hence $|a| - |b| \leq |a - b|$; exchanging the roles of a and b yields the inequality with the absolute value. The same proof holds in any NVS with $|\cdot|$ replaced by $\|\cdot\|$.

Thus, if $x_n \rightarrow x$ strongly, then $\|x_n\| \rightarrow \|x\|$. The converse is false⁴. From a quantum perspective, strong convergence of physical (i.e., normalized) states $|\psi_n\rangle \rightarrow |\psi\rangle$ with $\| |\psi_n\rangle \| = 1$ guarantees that the limiting state $|\psi\rangle$ is also normalized, $\| |\psi\rangle \| = 1$: the total probability remains equal to 1.

This notion of convergence may seem obvious if one restricts attention to the normed space $\mathbb{R}^n$, i.e., the usual Euclidean space. But the infinite-dimensional case is less straightforward. In a function space such as $L^2(\mathbb{R})$, which is a NVS and even a Hilbert space, the strong convergence just described concerns the convergence of a sequence of functions ψ_n toward a limiting function ψ .

Real analysis already provides notions of convergence for such sequences of functions, but they are not the same. On one hand, there is pointwise (or simple) convergence: ψ_n converges pointwise to ψ if $\psi_n(x) \rightarrow \psi(x)$ for every x . On the other hand, there is uniform convergence: the deviation between the sequence and its limit is controlled globally over the entire domain, i.e., $\sup_{x \in \mathbb{R}} |\psi_n(x) - \psi(x)| \rightarrow 0$. That is, for a given error tolerance ε , one can find a rank N beyond which the entire sequence falls within a tube of width 2ε (in modulus) around the limiting function. Uniform convergence clearly implies pointwise convergence.

Where does strong convergence fit in this picture? It is different from both. It is mean-square convergence. Indeed, in this setting it is expressed via the standard inner product of $L^2(\mathbb{R})$, and hence via the integral of the squared modulus:

$$\lim_{n \rightarrow \infty} \|\psi_n - \psi\|_{L^2} = 0 \iff \lim_{n \rightarrow \infty} \int_{-\infty}^{+\infty} |\psi_n(x) - \psi(x)|^2 dx = 0.$$

Strong convergence implies neither pointwise nor uniform convergence. Similarly, neither pointwise nor uniform convergence implies strong convergence in $L^2(\mathbb{R})$ ⁵.

Let us exhibit a counterexample showing that pointwise convergence does not imply strong convergence: consider the sequence of functions ψ_n on $[0, 1]$ defined by a peak of width $1/n$ and height $\sqrt{n}$:

$$\psi_n(x) = \begin{cases} \sqrt{n} & \text{if } 0 < x < \frac{1}{n} \\ 0 & \text{elsewhere} \end{cases}$$

Intuitively, this is a wave function one tries to localize more and more precisely. For any $x > 0$, there exists n large enough so that $\psi_n(x) = 0$. Therefore ψ_n converges pointwise to the zero function. However, computing the L^2 norm:

$$\|\psi_n\|_{L^2}^2 = \int_0^{1/n} n dx = 1$$

The L^2 norm remains constantly equal to 1, so the sequence cannot converge strongly to the zero function.

There is therefore a clear incompatibility between these two modes of convergence. This example shows that if pointwise convergence were the appropriate notion in quantum

⁴An immediate counterexample: $x_n = e_1$ if n is even, $x_n = e_2$ if n is odd, in an orthonormal basis. The norm is constant and equal to 1, but the sequence oscillates.

⁵On a finite interval, however, for example in $L^2([0, 1])$, uniform convergence does imply strong convergence.

mechanics, one could construct sequences of wave functions converging pointwise to zero, meaning that in the limit $n \rightarrow \infty$ the particle disappears (total probability zero), thereby violating the conservation of probability that is essential to the interpretation of the quantum formalism.

We will explain below how the quantum postulates actually impose the strong topology on the Hilbert space, see Section 3.10.

3.4.2 Weak convergence of a sequence of vectors

The reason we speak of strong topology and strong convergence is that a Hilbert space also admits a weak topology and a notion of weak convergence. The latter is convergence "up to an inner product":

Definition 3.5 : Weak convergence of a sequence of vectors

A sequence $(x_n)_{n \in \mathbb{N}}$ of elements of $\mathcal{H}$ converges weakly to $x \in \mathcal{H}$, denoted $x_n \rightharpoonup x$, if and only if

$$\forall y \in \mathcal{H}, \quad \langle y, x_n \rangle \rightarrow \langle y, x \rangle \quad (\text{convergence in } \mathbb{C}).$$

The interpretation is as follows. The inner product $\langle y, x_n \rangle$ represents the orthogonal projection of x_n onto the direction y . Weak convergence therefore requires each of these projections to converge. In quantum mechanics, $|\psi_n\rangle \rightharpoonup |\psi\rangle$ means that each probability amplitude $\langle \varphi | \psi_n \rangle$ converges individually to $\langle \varphi | \psi \rangle$ for all $|\varphi\rangle$. However, this coordinate-wise convergence does not impose strong convergence of the sequence. We have the following proposition:

Proposition 3.2 : Strong implies weak

Strong convergence implies weak convergence in every Hilbert space. The converse is false in infinite dimension.

Proof. We give here the canonical counterexample. Let $(e_n)_{n \in \mathbb{N}}$ be an orthonormal basis of $\mathcal{H}$, viewed as a sequence. It cannot converge strongly since each term is orthogonal to all others. Specifically, $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$: the vectors remain at constant mutual distance, so the sequence is not Cauchy and hence does not converge strongly (see the next section). Yet for any fixed $y \in \mathcal{H}$, Bessel's inequality, already encountered⁶ in Lesson 1, guarantees that $\langle y, e_n \rangle \rightarrow 0$. Thus an orthonormal basis of a Hilbert space, viewed as a sequence, converges weakly to zero but does not converge strongly. $\square$

A more physically meaningful counterexample is that of probability dilution. Suppose that $(|e_k\rangle)$ is an orthonormal basis of eigenstates of the Hamiltonian (for example, the energy levels of the harmonic oscillator). Consider the sequence of states:

$$|\psi_n\rangle = \frac{1}{\sqrt{n}} \sum_{k=1}^n |e_k\rangle$$

⁶For any orthonormal sequence (e_n) in a pre-Hilbert space, Bessel's inequality states that $\sum_{n=0}^{\infty} |\langle y, e_n \rangle|^2 \leq \|y\|^2 < +\infty$. Convergence of this series forces its general term to tend to zero.

Each term $|\psi_n\rangle$ represents an equiprobable superposition of the first n energy levels. As n increases, this sequence converges weakly to zero. Indeed, the probability amplitude onto any fixed state $|e_k\rangle$ is $\langle e_k|\psi_n\rangle = 1/\sqrt{n}$, which tends to zero as $n \rightarrow \infty$.

Yet the particle does not "disappear": the total probability, equal to the norm of the state, remains constant since $\|\psi_n\|^2 = \sum_{k=1}^n \frac{1}{n} = 1$. Here, the probability dilutes over an infinite number of states without ever concentrating on any of them. Since $\|\psi_n - 0\| = 1$ for all n , the sequence does not converge strongly to the zero vector. In fact, this sequence does not converge strongly to any element of $\mathcal{H}$, as it is not Cauchy: the states drift apart in the Hilbert space rather than stabilizing.

The continuous analog of the previous example consists of wave functions that spread out more and more in $L^2(\mathbb{R})$. The Gaussians

$$\psi_n(x) = \left(\frac{1}{\pi n}\right)^{1/4} e^{-x^2/(2n)}$$

of increasing width converge weakly to the zero function. Indeed, for any fixed $\varphi \in L^2(\mathbb{R})$, the integral $\int \varphi(x)\psi_n(x) dx$ tends to zero as ψ_n spreads out more and more. However, they remain normalized: $\|\psi_n\|_{L^2} = 1$ for all n , so $\|\psi_n - 0\|_{L^2} = 1 \not\rightarrow 0$ and the sequence does not converge strongly to zero.

All three counterexamples relied on the infinite dimension of the space. This was necessary: in finite dimension, weak convergence is equivalent to strong convergence.

To go further : Weak topology

Weak convergence arises from a genuine topology on $\mathcal{H}$, the *weak topology* $\mathcal{T}_f$, defined as the coarsest topology (i.e., with as few open sets as possible) making all maps $\ell_y : x \mapsto \langle y, x \rangle$, $y \in \mathcal{H}$, continuous. This topological definition is consistent with the sequential definition given above: we admit the equivalence

$$x_n \xrightarrow{\mathcal{T}_f} x \iff \forall y \in \mathcal{H}, \ell_y(x_n) \rightarrow \ell_y(x),$$

that is, $x_n \xrightarrow{\mathcal{T}_f} x \iff x_n \rightharpoonup x$. Here, $x_n \xrightarrow{\mathcal{T}_f} x$ denotes convergence in the general topological sense (every neighborhood of x in $\mathcal{T}_f$ contains x_n for all sufficiently large n), and not the metric convergence $\|x_n - x\| \rightarrow 0$ seen earlier.

The strong topology also makes the ℓ_y continuous (see the next section), so that the weak topology is, in infinite dimension, strictly coarser than the strong topology. It has fewer open sets, and hence more sequences converge. This is why strong convergence implies weak convergence but not conversely.

3.5 Cauchy sequences, completeness, and series

The two preceding notions of convergence assume that the limit x is known in advance. In practice, however, one often wishes to establish convergence of a sequence *without* an explicit candidate for the limit. It is therefore desirable to have an *intrinsic* convergence criterion. This is especially useful when asking whether an infinite sum converges. This is the purpose of Cauchy sequences and the resulting notion of completeness.

3.5.1 Cauchy sequences and completeness

In a normed vector space E , a sequence $(x_n)_{n \in \mathbb{N}}$ is called a *Cauchy sequence* if its terms become arbitrarily close to one another:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall p, q \geq N, \quad \|x_p - x_q\| < \varepsilon.$$

Cauchy sequences generalize the notion of strong convergence. They are precisely all sequences that could potentially converge, since every strongly convergent sequence is Cauchy: if $x_n \rightarrow x$ strongly, then the terms approach x and hence approach one another. By contrapositive, a sequence that is not Cauchy cannot converge strongly (this was used in the previous section). This provides a very useful criterion for showing that a sequence does not converge: it suffices to verify that its terms do not approach one another.

We thus have the formal implication $\text{strong} \Rightarrow \text{Cauchy}$, but the converse does not always hold in an arbitrary normed vector space. For this to be the case, the space must be *complete*, meaning it contains all the limits toward which Cauchy sequences converge.

Definition 3.6 : Completeness

A normed vector space E is said to be *complete* if and only if every Cauchy sequence converges in E :

$$\forall (x_n) \text{ Cauchy in } E, \quad \exists x \in E \text{ such that } x_n \rightarrow x.$$

Recall that a Hilbert space is, by definition, a complete pre-Hilbert space, see Lesson 1. Throughout the rest of this course, we will always assume we are working in a complete space. In this case, every convergent sequence is Cauchy and conversely.

3.5.2 Convergence of sums of vectors

A direct application of completeness concerns infinite sums of vectors, called *series*, which arise frequently in infinite-dimensional quantum mechanics. As with numerical series, convergence is defined via partial sums:

Definition 3.7 : Convergent series

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of elements of $\mathcal{H}$. The series $\sum_{n=0}^{+\infty} x_n$ is said to converge (strongly) to $S \in \mathcal{H}$ if and only if the sequence of partial sums

$$S_N = \sum_{n=0}^N x_n$$

converges strongly to S in $\mathcal{H}$, i.e., $\|S_N - S\| \rightarrow 0$ as $N \rightarrow +\infty$. One then writes

$$S = \sum_{n=0}^{+\infty} x_n.$$

There are two practical criteria for establishing convergence of a series. The first method is to determine whether the series of norms itself converges.

Proposition 3.3 : Absolute convergence

If the numerical series $\sum_{n=0}^{+\infty} \|x_n\|$ converges in $\mathbb{R}$, the series is said to converge *absolutely*. Absolute convergence then implies that the series $\sum_{n=0}^{+\infty} x_n$ converges in $\mathcal{H}$.

This approach is "crude" in the sense of being overly demanding. It does not exploit possible cancellations between the terms x_n that might stabilize the sum. The converse is therefore false: a series can converge without the sum of its individual norms ($\|x_n\|$) being finite. The other method is to apply the Cauchy criterion directly to the partial sums, yielding the following criterion:

Proposition 3.4 : Cauchy criterion

The series $\sum_{n=0}^{+\infty} x_n$ converges in $\mathcal{H}$ if and only if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall q > p \geq N, \quad \left\| \sum_{n=p}^q x_n \right\| < \varepsilon.$$

The Cauchy criterion is sharper since it concerns the norm of partial sums, not the sum of norms.

Example 3.1 : Convergence of series

Consider first the series with $x_n = \frac{1}{n^2}e_n$ in a Hilbert basis (e_n) . It is *absolutely convergent* since the series of norms $\sum \frac{1}{n^2}$ converges.

Now consider the series with $x_n = \frac{(-1)^n}{n}e_1$ (which stays on the e_1 axis). It is not absolutely convergent since $\sum \frac{1}{n}$ diverges. It nevertheless converges by the Cauchy criterion through partial cancellation of successive terms of opposite sign: factoring out e_1 , it is a classical result for real alternating series that the series converges to $-\ln(2)$. Thus:

$$\sum_{n=1}^{+\infty} \frac{(-1)^n}{n} e_1 = -\ln(2)e_1$$

Finally, the case of the series with $x_n = \frac{1}{n}e_n$ is particularly interesting. It is also not absolutely convergent, yet it converges to a vector $x = (1, 1/2, 1/3, \dots)$ in $\ell^2(\mathbb{N})$. Here, stability comes not from alternating signs but from the pairwise orthogonality of the terms. The Pythagorean theorem indeed gives, in the Cauchy criterion:

$$\left\| \sum_{n=p}^q \frac{1}{n} e_n \right\|^2 = \sum_{n=p}^q \frac{1}{n^2}$$

Since the tail of the series $\sum \frac{1}{n^2}$ tends to 0, the Cauchy criterion is satisfied.

3.6 Continuity of maps

Before defining continuity, let us briefly recall the notion of limit in normed spaces.

3.6.1 Limit of a map

Definition 3.8 : Limit

Let E, F be two normed spaces and $f : \mathcal{D}(f) \subset E \rightarrow F$ a map defined on its domain $\mathcal{D}$. Let $x_0 \in \overline{\mathcal{D}(f)}$ and $\ell \in F$. We say that f admits ℓ as a *limit* at x_0 if

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall x \in \mathcal{D}(f), \|x - x_0\|_E < \delta \implies \|f(x) - \ell\|_F < \varepsilon.$$

One then writes $\lim_{x \rightarrow x_0} f(x) = \ell$. When it exists, the limit is unique.

Note that in the case $E = F = \mathbb{R}$, this recovers the usual definition of the limit of a real-valued function. In that setting, the order structure also allows one to define left and right limits. This is not possible in a general normed space: a point x_0 can be approached along infinitely many paths. One therefore requires that all sequences $x_n \rightarrow x_0$ yield the same limit for $f(x_n)$. Note also that the definition encompasses points in the closure of the domain, not just those belonging to the domain itself. This allows one to define, when it exists, a limit at the boundary of the domain, where the function is not necessarily defined. For example, this allows one to define the limit at zero of $f(x) = \sin(x)/x$, which equals 1.

This notion of limit can be reformulated in terms of sequences, which is often more tractable in practice:

Proposition 3.5 : Sequential characterization

In a normed space, the following equivalence holds:

$$\lim_{x \rightarrow x_0} f(x) = \ell \iff \forall (x_n) \subset \mathcal{D}(f), x_n \rightarrow x_0 \implies f(x_n) \rightarrow \ell.$$

in the sense of strong convergence, that is, $\|x_n - x_0\| \rightarrow 0 \implies \|f(x_n) - \ell\| \rightarrow 0$.

3.6.2 Continuity of a map at a point

The notion of continuity is closely related to that of limit. A map is said to be continuous at x_0 if the limit it admits is precisely $f(x_0)$.

Definition 3.9 : Continuity at a point

A map $f : \mathcal{D}(f) \subset E \rightarrow F$ is *continuous at* $x_0 \in \mathcal{D}(f)$ if it admits a limit at x_0 equal to $f(x_0)$:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

In ε - δ form, this definition reads:

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall x \in \mathcal{D}(f), \|x - x_0\|_E < \delta \implies \|f(x) - f(x_0)\|_F < \varepsilon.$$

Note that continuity at x_0 is only defined for $x_0 \in \mathcal{D}(f)$. If $x_0 \in \overline{\mathcal{D}(f)} \setminus \mathcal{D}(f)$ and f admits a limit ℓ at x_0 , one can define the *continuous extension* of f at x_0 by setting $\tilde{f}(x_0) = \ell$. Such an extension is unique when it exists.

A map is said to be continuous if it is continuous at every point of its domain.

3.6.3 The case of linear maps

When the map is linear, one more commonly speaks of a *linear operator* rather than a map, but the two are synonymous. We will denote it T . This case is absolutely central in quantum mechanics, which is a linear theory.

The question of continuity is radically simplified in this case. The key idea is to exploit the identity $T(x + a) = T(x) + T(a)$ to reduce the question of continuity at an arbitrary point to continuity at the origin. This yields one of the main theorems of this chapter:

Theorem 3.2 : Global continuity criterion for linear maps

Let $T : \mathcal{D}(T) \subset E \rightarrow F$ be a linear operator between two normed spaces. The following statements are equivalent:

1. T is continuous at some point $x_0 \in \mathcal{D}(T)$
2. T is continuous at 0 (note that $T(0) = 0$ by linearity)
3. T is continuous on all of $\mathcal{D}(T)$

For a linear operator, continuity is therefore a global property: either it is continuous everywhere on its domain, or nowhere on it.

Proof. (1) $\Rightarrow$ (2): If T is continuous at x_0 , then $\|T(x_0 + h) - T(x_0)\|_F \rightarrow 0$ as $\|h\|_E \rightarrow 0$. By linearity, $T(x_0 + h) - T(x_0) = T(h)$, so $\|T(h)\|_F \rightarrow 0$: T is continuous at 0.

(2) $\Rightarrow$ (3): For any $x_0 \in \mathcal{D}(T)$ and $x \in \mathcal{D}(T)$, linearity gives $T(x) - T(x_0) = T(x - x_0)$. If $x \rightarrow x_0$, then $x - x_0 \rightarrow 0$, and by continuity at 0, $T(x - x_0) \rightarrow 0$, hence $T(x) \rightarrow T(x_0)$.

(3) $\Rightarrow$ (1): Immediate. $\square$

Another major proposition of this chapter (often stated as part of the theorem above) concerns the link between continuity and boundedness of a linear map (for a general definition of bounded maps, see the next section).

Proposition 3.6 : Continuity and boundedness

Let $T : E \rightarrow F$ be a linear operator between normed spaces. Then T is continuous at 0 if and only if there exists a constant $C \geq 0$ such that

$$\|T(x)\|_F \leq C\|x\|_E \quad \text{for all } x \in E,$$

in which case T is said to be bounded.

Proof. ($\Leftarrow$) If such a constant C exists, then for any $\varepsilon > 0$, it suffices to take $\delta = \varepsilon/C$ (or any δ if $C = 0$): whenever $\|x\|_E < \delta$, one has $\|T(x)\|_F \leq C\|x\|_E < C\delta = \varepsilon$, which gives continuity at 0.

($\Rightarrow$) Suppose T is continuous at 0. Applying the definition with $\varepsilon = 1$, there exists $\delta > 0$ such that $\|x\|_E < \delta$ implies $\|T(x)\|_F < 1$. The idea is then to exploit linearity as a scaling. Let $x \in E$ be nonzero. The vector $u = \frac{\delta}{2} \cdot \frac{x}{\|x\|_E}$ satisfies $\|u\|_E = \delta/2 < \delta$, hence $\|T(u)\|_F < 1$. By linearity, the scaling factor propagates faithfully through T :

$$\|T(x)\|_F = \frac{2\|x\|_E}{\delta} \|T(u)\|_F < \frac{2}{\delta} \|x\|_E.$$

The constant $C = 2/\delta$ works (the inequality being trivially satisfied for $x = 0$). $\square$

This characterization allows one in particular to prove the following corollary:

Corollary 3.1 : Continuity in finite dimension

Every linear operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous.

It suffices to show that it is bounded, which follows almost immediately.

Proof. Let $(e_1, \dots, e_n)$ denote the canonical basis of $\mathbb{R}^n$. For any $x = \sum_{i=1}^n x_i e_i$, the triangle inequality and linearity give

$$\|T(x)\|_F \leq \sum_{i=1}^n |x_i| \|T(e_i)\|_F \leq \left(\sum_{i=1}^n \|T(e_i)\|_F \right) \max_{1 \leq i \leq n} |x_i| \leq C \|x\|_\infty,$$

where $C = \sum_{i=1}^n \|T(e_i)\|_F$ is a finite constant. Since the norm $\|\cdot\|_\infty$ is equivalent to any norm on $\mathbb{R}^n$, the result holds for any choice of norm on the domain. $\square$

3.6.4 Discontinuity and infinite dimension

In infinite dimension, a linear operator can be either continuous or discontinuous, and if it is discontinuous, it is discontinuous everywhere on its domain. This is a rather counterintuitive object. One may legitimately ask what this means, especially if one's intuition of discontinuity is limited to real-valued functions, which have "jumps" with different left and right limits. Does an everywhere-discontinuous operator have jumps everywhere? The answer is no. In fact, the nature of discontinuity in these two cases is completely different.

To understand this better, let us first negate the ε - δ formulation of continuity at 0:

$$\exists \varepsilon > 0, \forall \delta > 0 : \exists x \in E : \|x\|_E < \delta \text{ and } \|T(x)\|_F \geq \varepsilon.$$

In other words, discontinuity at zero requires the existence of a threshold ε such that for every radius δ , however small, there exists a vector x_δ in that small ball whose image "resists," i.e., does not drop below the threshold ε .

One might naively suppose that discontinuity arises from a fixed direction u along which the action of T blows up. But this is impossible: for any fixed vector $u \in \mathcal{D}(T)$, $T(u)$ is a vector in F , and its norm $\|T(u)\|_F$ is necessarily finite. By linearity, taking a sufficiently small input vector makes its image arbitrarily small. Thus the action (often called the amplification) of the operator is controlled in each individual direction.

What actually happens is more subtle. It is a sequential phenomenon: in the presence of discontinuity, one can find a *sequence* of unit directions (u_n) whose images escape to infinity. To see this, fix a threshold ε and choose a sequence of radii $\delta_n = 1/n$ tending to zero. Discontinuity guarantees the existence of vectors x_n (with $\|x_n\| \leq 1/n$) such that $\|T(x_n)\| \geq \varepsilon$. Setting $u_n = \frac{x_n}{\|x_n\|}$, linearity gives:

$$\|T(u_n)\| = \frac{\|T(x_n)\|}{\|x_n\|} \geq n\varepsilon \xrightarrow{n \rightarrow \infty} \infty$$

Discontinuity thus manifests as an *escape* in the space of directions: no fixed direction is problematic, but one can find a sequence of directions along which the action of the operator diverges. This is why the phenomenon can only occur in infinite dimension.

Example 3.2 : The differentiation operator is discontinuous

The standard example is the differentiation operator $D : f \mapsto f'$, defined on a dense domain of $L^2([0, 1])$ (the differentiable functions with square-integrable derivative). We admit that this space has the trigonometric functions $e_n(x) = \sqrt{2} \sin(n\pi x)$ as a Hilbert basis, each with norm $\|e_n\|_{L^2} = 1$. One computes:

$$D(e_n)(x) = n\pi\sqrt{2}\cos(n\pi x), \quad \|D(e_n)\|_{L^2} = n\pi \xrightarrow{n \rightarrow \infty} +\infty.$$

This exhibits the escape described above: no individual e_n is problematic ($D(e_n)$ remains in L^2), but the *sequence* (e_n) explores increasingly high frequencies, which, through differentiation, leads to a blow-up of the amplification.

Let us return to the geometric interpretation. Can one still speak of a jump? For the function from $\mathbb{R}$ to $\mathbb{R}$ given by

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

the discontinuity at 0 is revealed by a *single* fixed approach path $x \rightarrow 0^-$, which shows that $f(0^-) \neq f(0)$ and hence reveals a jump.

In the case of the differentiation operator near a given point ψ , no *fixed* approach path reveals the discontinuity, since for any fixed direction $u \in \mathcal{D}(D)$, one has $D(\psi + \lambda u) = \psi' + \lambda u' \rightarrow \psi'$ as $\lambda \rightarrow 0$: there is no jump along u , for any function u .

On the other hand, the sequence $f_n = \psi + \frac{1}{\sqrt{n}} \sin(n \cdot)$ approaches ψ through ever-changing directions: one has $f_n \rightarrow \psi$ in the L^2 norm, but $f'_n \not\rightarrow \psi'$, due to the divergence noted in the example above.

Important point 3.1

Discontinuity in infinite dimension thus reveals itself only through this *mobility of the approach direction*, while remaining *invisible direction by direction*. In this sense, it is more a global phenomenon on the space of directions than a localized jump: it reflects the impossibility of uniformly controlling the amplification of an operator on the unit sphere.

Remark: In point-particle quantum mechanics, the momentum operator is represented by $i\hbar D$, and is therefore discontinuous. As a consequence, it can only be defined on a strict subdomain of L^2 . Its self-adjointness is therefore not automatic, and depends on the boundary conditions imposed on the wave functions. This will all be detailed in the following lessons.

3.6.5 The case of linear functionals

In the previous lesson we saw the fundamental role of linear functionals on a Hilbert space, and in particular that of continuous ones. For completeness, we now detail what

these continuous functionals are. We first have the following proposition:

Proposition 3.7 : Continuity of the functionals ℓ_y

Let $\mathcal{H}$ be a complex Hilbert space equipped with a Hermitian inner product $\langle \cdot, \cdot \rangle$ and the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$. For every $y \in \mathcal{H}$, the linear functional

$$\ell_y : \mathcal{H} \rightarrow \mathbb{C}, \quad x \mapsto \langle y, x \rangle$$

is continuous.

Proof. The Cauchy-Schwarz inequality gives, for all $x \in \mathcal{H}$,

$$|\ell_y(x)| = |\langle y, x \rangle| \leq \|y\| \cdot \|x\|,$$

which shows that ℓ_y is bounded, hence continuous (Proposition 3.6). (Note: this statement is in fact valid on any pre-Hilbert space over $\mathbb{R}$ or $\mathbb{C}$.) $\square$

Moreover, recall that the Riesz representation theorem states that the converse holds: every continuous linear functional on $\mathcal{H}$ is of this form. The map $y \mapsto \ell_y$ is therefore an isometric bijection from $\mathcal{H}$ onto its topological dual $\mathcal{H}^*$, and this is what justifies the identification of vectors (kets) with continuous linear functionals (bras).

3.6.6 Continuity of the inner product

The Cauchy-Schwarz inequality also yields continuity of the inner product, which is not itself a linear map in the strict sense, but rather sesquilinear: antilinear in the first argument and linear in the second. Without going into details, we ask about *joint* continuity in both arguments. This indeed holds: for pairs $(x_n, y_n) \rightarrow (x, y)$ in $\mathcal{H} \times \mathcal{H}$, one has $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$. Proof:

$$|\langle y_n, x_n \rangle - \langle y, x \rangle| = |\langle y_n - y, x_n \rangle + \langle y, x_n - x \rangle| \leq \|y_n - y\| \|x_n\| + \|y\| \|x_n - x\| \rightarrow 0.$$

We used the triangle inequality, applied Cauchy-Schwarz twice, and used the fact that $\|x_n\|$ remains bounded to conclude convergence to zero.

Remark: As already noted in Lesson 1, the continuity of linear functionals and of the inner product expresses physically the robustness of probability amplitudes under small perturbations of states, thereby ensuring the stability of measured probabilities.

3.7 Topology of vector subspaces

In what follows, we will frequently work with vector subspaces of normed spaces or Hilbert spaces. When discussing their topology, we implicitly mean the *induced topology*: a subspace $F \subset E$ naturally inherits the norm of E by restriction, and hence the notions of open set, closed set, convergence, etc. Whether these subspaces are closed or dense depends critically on dimension. We first have the following theorem:

Theorem 3.3

In a normed vector space E , every finite-dimensional vector subspace is closed.

In particular, they can never be dense, unless they coincide with the whole space (the trivial case). The case of infinite-dimensional subspaces is richer: some are closed and others are not.

Density and infinite dimension If E is infinite-dimensional, a vector subspace F of E can be dense without being equal to E . This is a counterintuitive phenomenon, since in finite dimension a proper subspace always lacks at least one direction to fill the whole space. In infinite dimension, however, the following holds:

Proposition 3.8

Let F be a vector subspace of a normed space E . If F is dense in E and $F \neq E$, then F is necessarily infinite-dimensional.

The proof is immediate: if F were finite-dimensional, it would be closed by the preceding theorem. Since F is dense, this would give $E = \overline{F} = F$, contradicting the assumption.

It is precisely because it has infinitely many directions that a subspace can "spread" throughout the whole space (be dense) without entirely filling it (without being closed).

Example: finitely supported sequences Consider the example of c_{00} , the set of finitely supported sequences. This is a subspace of $\ell^2(\mathbb{N})$, the space of square-summable sequences. The expansion of each element $x \in c_{00}$ uses only finitely many vectors of the canonical basis $(e_n)_{n \in \mathbb{N}}$, but this number is arbitrary: there is no uniform bound on the number of directions used. This is also why this subspace is closed under addition: the sum of two elements of c_{00} has finite support, possibly with a larger support. Furthermore, this shows it is dense, since any point $x = (x_n)_{n \in \mathbb{N}} \in \ell^2(\mathbb{N})$ can be approximated by the sequence of elements of c_{00} obtained by truncating x at its N -th term:

$$x^{(N)} = (x_0, x_1, \dots, x_{N-1}, 0, 0, \dots) \in c_{00}$$

which converges strongly to x as $N \rightarrow \infty$.

This subspace is therefore dense because it is free to use any available direction, in an arbitrarily large number, which allows one to approximate any point of the full space.

The family $\{e_n\}_{n \in \mathbb{N}}$ plays a dual role: it is both an algebraic basis of c_{00} (every element of c_{00} is a finite linear combination of the e_n) and a Hilbert basis of ℓ^2 (every element of ℓ^2 is the limit of a series involving the e_n).

Example: continuous functions However, the mechanism of density is not always of the "growing finite support" type. Indeed, let $C([0, 1])$ denote the set of continuous functions on $[0, 1]$. This is a vector subspace of $L^2([0, 1])$. One can show it is dense by first approximating any L^2 function by step functions, and then these by continuous functions.

Unlike the case of c_{00} , however, the algebraic nature of this density remains elusive. In the first case, the density could be explained algebraically: the (algebraic) basis of c_{00} contains all the vectors of the Hilbert basis e_n , and density uses "all directions" of ℓ^2 , with no uniform bound on the number of directions simultaneously mobilized.

For $C([0, 1])$ dense in $L^2([0, 1])$, no such statement can be made. A general result⁷ does affirm that $C([0, 1])$ possesses an algebraic basis, but:

1. No explicit or constructive description of it is known,
2. The density is proved by analytic arguments and escapes any purely algebraic understanding.

In conclusion, dense subspaces in infinite dimension are subtle, and sometimes surprising, objects. Unlike the finite-dimensional case where a proper subspace can never be dense, infinite dimension allows this phenomenon, and it is not always possible to express it in purely algebraic terms.

In quantum mechanics, unbounded (or discontinuous) linear operators, such as the momentum operator encountered earlier, can only be defined on a proper vector subspace of L^2 . In order for them to still qualify as observables, one requires this subspace to be dense: this guarantees that no physical state is topologically inaccessible from the domain of the operator, and it is also the condition for the adjoint of the operator to be uniquely defined (see the next lesson).

3.8 Bounded sets

In a normed space $(E, \|\cdot\|)$, a subset A is said to be **bounded** if there exists $M > 0$ such that

$$\|x\| \leq M \quad \text{for all } x \in A.$$

Unboundedness has a sequential characterization: A is not bounded if and only if there exists a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of A such that $\|x_n\| \rightarrow +\infty$.

As seen in Section 3.6.3, it is the notion of bounded set that characterizes continuity of linear maps between normed spaces: a linear map $T : E \rightarrow F$ is continuous if and only if it is bounded, meaning it maps every bounded subset of E to a bounded subset of F .

For the record, note that bounded and closed are two independent notions:

- The open ball $B(0, 1) = \{x \in E : \|x\| < 1\}$ is bounded but not closed.
- A linear line $\mathbb{R} \cdot v$ (with $v \neq 0$) is closed but not bounded.
- The closed ball $\overline{B}(0, r)$ of radius $r > 0$ is both bounded and closed.

3.9 Compact sets

Compactness has a purely topological definition that we will not detail here. In normed spaces it admits a sequential characterization (the Bolzano-Weierstrass theorem) that we will take as a definition:

⁷Every vector space possesses an algebraic basis. This statement relies on the axiom of choice and is proved via Zorn's lemma.

Definition 3.10 : Compact set; precompact set

Let $(E, \|\cdot\|)$ be a normed space.

- A subset $K \subset E$ is called **compact** if every sequence of elements of K admits a subsequence converging to an element of K .
- A subset $B \subset E$ is called **precompact** (or **relatively compact**) if its closure $\overline{B}$ is compact.

A precompact set is thus nearly compact: it suffices to adjoin its limit points to make it closed and compact. A compact set is necessarily closed and bounded⁸. The Heine-Borel theorem tells us that the converse holds in finite dimension: a set is compact if and only if it is closed and bounded⁹.

We sketch the proof in finite dimension, as it explains why the converse fails in infinite dimension. On a closed interval $[a, b]$ of $\mathbb{R}$, given an infinite sequence of points, one can bisect the interval: at least one half contains infinitely many points. Iterating this bisection, one constructs a sequence of nested intervals whose length tends to zero and each of which contains infinitely many points of the sequence. This allows one to extract a subsequence converging to the unique point common to all these intervals.

In dimension n , one generalizes this process coordinate by coordinate: one first extracts a subsequence converging in the first coordinate, then from it a further subsequence converging in the second, and so on. Since there are only finitely many directions to handle, one always ends up with a convergent subsequence, see Figure 3.3.

In infinite dimension, boundedness allows one to apply the bisection argument detailed above, but this does not control the infinitely many directions along which points of the sequence can escape. We thus have the striking counterexample due to Riesz:

Theorem 3.4 : Riesz

In an infinite-dimensional NVS, the closed unit ball $\overline{B}(0, 1)$ is not compact.

The proof is not immediate in the case of a general NVS, but is very simple in a Hilbert space. It suffices to consider the sequence $(e_n)_{n \in \mathbb{N}}$ of vectors of a Hilbert basis (which is orthonormal). One has $\|e_n - e_m\| = \sqrt{2}$ for all $n \neq m$. Since the points remain at constant mutual distance, no subsequence can be Cauchy, and hence none can converge.

The unit ball seems compact at first glance because one tends to picture it in finite dimension, but in reality it is too large in infinite dimension to be compact. In contrast, a much smaller set, such as the Hilbert cube:

$$H = \{(x_n)_{n \in \mathbb{N}} \in \ell^2 \mid |x_n| \leq \frac{1}{n}\}$$

is compact because it becomes arbitrarily thin in the high-index directions, which forces the existence of accumulation points.

⁸The argument is simple: if it were not bounded, there would exist a sequence escaping to infinity with no convergent subsequence. If it were not closed, a sequence could converge to a point outside the set, and every subsequence would converge to that same point, contradicting the definition.

⁹In particular, the compact subsets of $\mathbb{R}$ are exactly the closed intervals $[a, b]$ and finite sets (for example $\{1, 5, 12\}$).

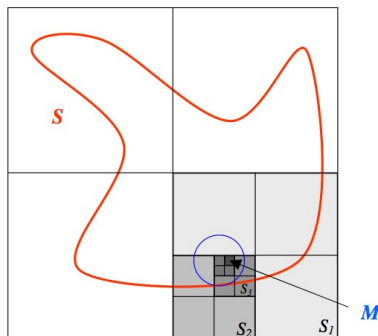


Figure 3.3: A compact set S in a 2-dimensional NVS. One begins by enclosing it in a rectangle, which is then divided into four equal parts; the procedure is repeated by always choosing a sub-rectangle that still contains infinitely many points of the sequence. Here we do not proceed coordinate by coordinate but, equivalently, by successive bisections of nested rectangles. This constructs an accumulation point M that serves as the limit of an extracted subsequence (the extraction is always possible since, by construction, infinitely many points of the sequence remain).

<https://old.maa.org/press/periodicals/convergence/an-analysis-of-the-first-proofs-of-the-heine-borel-theorem-cousins-proof>

To go further : Weak compactness

In a Hilbert space, the Banach-Alaoglu theorem guarantees that the closed unit ball is *weakly compact*: every bounded sequence admits a subsequence that converges weakly (i.e., such that $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ for all $y \in E$).

We saw that the closed unit ball cannot be strongly compact because it contains the sequence (e_n) of an orthonormal basis, whose vectors remain at constant mutual distance $\sqrt{2}$. However, this counterexample no longer applies in the context of weak convergence. Indeed, observing this sequence along any fixed direction y , its projection $\langle y, e_n \rangle$ necessarily tends to 0 by Bessel's inequality. Thus what prevented strong convergence (the fact that the vectors escape into orthogonal dimensions) is no longer an obstacle here: the sequence (e_n) does converge weakly to the origin.

These considerations may seem abstract, but this result is needed to prove an important theorem in quantum mechanics: every quantum system with a Hamiltonian bounded from below and subject to a confining potential (i.e., one that diverges at infinity) always admits a ground state.

Here is the idea of the proof. Physical states are normalized and thus belong to the closed unit ball. The weak compactness of the latter guarantees that from any sequence of physical states with decreasing energies, one can extract a weakly convergent subsequence. The confining potential then prevents the wave

function from escaping to (spatial) infinity, ensuring that the norm of the limit is also equal to 1^a. Weak convergence together with convergence of the norm implies strong convergence, see Theorem 3.5 below. One concludes that this sequence of states converges to a physical state minimizing the mean energy: this is the ground state. See details in [Reed and Simon (Vol. 4), Theorem XIII.1].

^aThis is the delicate part of the proof, which we will not detail here.

3.10 The appropriate topology in quantum mechanics

All the preceding sections have prepared us for the question of the appropriate topology to use in quantum mechanics. Consider a sequence¹⁰ of physical (normalized) states $|\psi_n\rangle$ converging to a limiting state $|\psi\rangle$. It is natural to require that experimental predictions converge consistently: measurement probabilities $P_n \rightarrow P$ and mean values of observables $\langle \hat{A} \rangle_n \rightarrow \langle \hat{A} \rangle$. The question is: in what sense should the convergence $|\psi_n\rangle \rightarrow |\psi\rangle$ be understood? That is, for which topology?

The measurement postulate states that the probability of obtaining the outcome $|\varphi\rangle$ when the system is prepared in state $|\psi\rangle$ is given by: $P(\psi \rightarrow \varphi) = |\langle \varphi | \psi \rangle|^2$. For the probabilities to converge, it is therefore sufficient that the probability amplitudes converge:

$$\forall |\varphi\rangle \in \mathcal{H}, \quad \langle \varphi | \psi_n \rangle \xrightarrow{n \rightarrow \infty} \langle \varphi | \psi \rangle,$$

which is nothing other than the weak convergence seen earlier: $|\psi_n\rangle \rightharpoonup |\psi\rangle$. Weak convergence also guarantees convergence of the mean values of bounded observables¹¹.

The weak topology thus seems like a good candidate. But, as seen in the counterexamples from the previous sections, weak convergence is not sufficient to impose strong convergence. In particular, the norm does not necessarily converge:

$$|\psi_n\rangle \rightharpoonup |\psi\rangle \not\Rightarrow \|\psi_n\| \rightarrow \|\psi\|.$$

In quantum mechanics, however, conservation of total probability is *mandatory* for the interpretation of the theory. One therefore requires convergence of the norm: $\|\psi_n\| \rightarrow \|\psi\|$, which physically means there is no probability leakage. The following theorem allows one to combine these two requirements:

Theorem 3.5 : Criterion for strong convergence

In a Hilbert space, the sequence (x_n) converges strongly to x if and only if it converges weakly to x and converges in norm:

$$x_n \rightarrow x \text{ (strongly)} \iff x_n \rightharpoonup x \text{ and } \|x_n\| \rightarrow \|x\|$$

We arrive at the following remarkable conclusion, which closes this lesson: the strong topology is not merely natural from a mathematical standpoint, it is *imposed* in quantum mechanics by the consistency of the postulates. First, because probabilities are expressed

¹⁰Since every state in the Hilbert space is assumed to represent a genuine physical state, one can always contemplate such sequences. It does not matter whether one can physically realize them (they involve infinitely many states): the point is that they can in principle be considered.

¹¹This is slightly more technical and will be revisited later; we admit it for now (Section (X)).

via continuous linear functionals (weak convergence), and second, because of conservation of total probability (convergence of the norm).